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Corporate Bond Liquidity

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Contents

1	Introduction	3
2	Multicurve Bootstrap	3
2.a	Methodology	3
2.b	OIS Bootstrap	4
2.c	Euribor 6m Bootstrap	4
2.d	Multi-curve bootstrap results	5
3	Volatility Calibration	6
3.a	Market prices: the Bachelier (Normal-Black) formula	6
3.b	Model prices: the closed-form CS swaption	7
3.c	Results and interpretation	8
4	Bounds of the liquidity premium	9
4.a	Methodology	9
4.b	Zeta Spread Bootstrap	9
4.c	Lower and upper bounds	10
5	Bond yields	12
6	Conclusion	15

1 Introduction

Fixed income markets distinguish between liquid and illiquid bonds issued by the same obligor. A liquid bond can be sold immediately at its fair market price, while an illiquid bond requires time to find a counterparty willing to buy it. This time-to-liquidate (TTL), denoted τ , creates an additional source of risk: if the issuer defaults before the bond is sold, the holder receives nothing under a zero-recovery assumption. Rational investors therefore demand a premium for holding illiquid bonds, known as the sheer liquidity premium Δ_τ .

We work in the EUR interbank market as of 10 September 2015. The market data - interest rate curves, swaption volatilities, and bond prices - are all sourced from this date.

The objective of this study is to quantify the sheer liquidity premium, and then compute bond yields for 2 banks (Santander and BNP Paribas). To achieve this, we will structure the project in the following way. First, we bootstrap two interest rate curves from EUR interbank instruments: the OIS discounting curve, constructed from EONIA deposits and OIS swaps, and the Euribor 6m pseudo-discount curve, constructed from Euribor deposit, FRAs, and interest rate swaps. These two curves are distinct objects in the post-2008 multicurve framework Baviera and Cassaro [2]. Second, we calibrate the volatility parameters of a parsimonious Hull-White model from diagonal cash-settled swaption prices quoted in normal (Bachelier) volatility Baviera [1]. Third, for each issuer we bootstrap a credit (Zeta-spread) curve from its liquid bonds and apply the closed-form upper and lower bounds of Baviera, Nassigh, and Nastasi [3] to compute the sheer liquidity premium Δ_τ for BNP Paribas and Santander bonds, for TTL values of two weeks and two months. Finally, we translate the liquidity premium from price space into yield space, reporting the yield spread between liquid and illiquid bonds for each issuer.

We find that the upper and lower bounds essentially coincide – their difference is of order 10^{-8} per unit face value – so the liquidity premium is determined clearly. Translated into yield space, it produces a small but strictly positive liquidity yield spread for both issuers, which widens with the time-to-liquidate (larger at two months than at two weeks).

2 Multicurve Bootstrap

2.a Methodology

Concerning the data, we will use EONIA deposits and OIS swaps to build the OIS curve $P^D(t_0, T)$, used for discounting. And we will use FRAs, IRS and one 6-month Euribor deposit to build the Euribor 6m pseudo-discount curve $P(t_0, T)$.

All dates are adjusted using the modified-following (mod-foll) convention. And concerning interpolation, in both bootstraps, we use linear interpolation on zero rates with an Act/365 time axis. The continuously compounded zero rate is defined as

$$z(t_0, T) = -\frac{\ln P(t_0, T)}{T - t_0} \quad (1)$$

Between two pillar dates $T_a < T < T_b$, the zero rate is interpolated linearly:

$$z(T) = z(T_a) + \frac{T - T_a}{T_b - T_a} (z(T_b) - z(T_a)) \quad (2)$$

and the discount factor recovered as $P(t_0, T) = e^{-z(T)(T-t_0)}$.

2.b OIS Bootstrap

The OIS curve provides 18 discount factors $P^D(t_0, T_i)$ for maturities $T_1, \dots, T_{18}$ from 1 week to 12 years. The bootstrap distinguishes two cases based on the total year fraction $\delta(t_0, T_i)$ in Act/360. For instruments with maturity up to and including 1 year, the discount factor is obtained directly from the deposit formula:

$$P^D(t_0, T_i) = \frac{1}{1 + \delta(t_0, T_i) \cdot R^{OIS}(t_0, T_i)} \quad (3)$$

where $\delta(t_0, T_i)$ is computed in Act/360 and R^{OIS} is the mid-quoted OIS rate. This applies to the first 6 instruments.

For higher maturities, the instrument is an OIS swap where the fixed rate $R^{OIS}(t_0, T_i)$ is exchanged annually against the daily compounded EONIA rate. We use the following recursive formula [2]:

$$P^D(t_0, T_i) = \frac{1 - R^{OIS}(t_0, T_i) \sum_{k=1}^{i-1} \delta_k P^D(t_0, T_k)}{1 + \delta_i \cdot R^{OIS}(t_0, T_i)} \quad (4)$$

where $\delta_k = \delta(T_{k-1}, T_k)$ is the Act/360 year fraction of the k -th annual period and the intermediate discount factors $P^D(t_0, T_k)$ are obtained by interpolating on the already-bootstrapped pillars.

2.c Euribor 6m Bootstrap

The Euribor 6m pseudo-discount curve $P(t_0, T)$ is bootstrapped following the forward-backward methodology of [2]. The curve knots are: t_0 , monthly knots $\hat{t}_1, \dots, \hat{t}_5, \hat{t}_6, \hat{t}_7, \dots, \hat{t}_{12}, \hat{t}_{18}$, and annual knots $t_{2y}, \dots, t_{12y}$, where $\hat{t}_k$ denotes the date k months after t_0 with mod-foll adjustment. The bootstrap proceeds in six steps, and will produce 24 pseudo-discount pillars.

First, the 6-month Euribor deposit rate L^{6m} gives the first pseudo-discount factor directly from the deposit formula (analogous to (3)).

Then, the FRA with start date $\hat{t}_6$ and end date $\hat{t}_{12}$ provides the forward rate $F_{6 \times 12}$. The pseudo-discount at 12 months is obtained by a forward move:

$$P(t_0, \hat{t}_{12}) = \frac{P(t_0, \hat{t}_6)}{1 + \delta(\hat{t}_6, \hat{t}_{12}) \cdot F_{6 \times 12}} \quad (5)$$

Now we have two anchor points $P(\hat{t}_6)$ and $P(\hat{t}_{12})$ established, we first **interpolate** to obtain $P(\hat{t}_7), \dots, P(\hat{t}_{11})$. These intermediate knots are necessary for the backward move.

We then use the FRAs $1 \times 7, 2 \times 8, 3 \times 9, 4 \times 10, 5 \times 11$ to recover $P(\hat{t}_1), \dots, P(\hat{t}_5)$ via a backward formula.

After that, we move to the 18 month and 2y swaps. The 18-month swap rate is not directly quoted; it is obtained by cubic spline interpolation. Using this, we obtain,

$$F_3 = \frac{I(1) - w_1 F_1 - w_2 F_2}{2w_3} \quad (6)$$

where $w_k = \tilde{\delta}_k \cdot P^D(t_0, \tilde{t}_k)$ are the floating-leg weights using OIS discount factors, F_1 and F_2 are the already-known forward rates from Steps 1 and 2, and $I(1)$ is the fixed-leg annuity of the 18m

swap. Then, From F_3 we recover:

$$P(t_0, \hat{t}_{18}) = \frac{P(t_0, \hat{t}_{12})}{1 + \delta(\hat{t}_{12}, \hat{t}_{18}) \cdot F_3} \quad (7)$$

We obtain after $F_4 = (I(2) - I(1))/w_4$, and $P(t_0, t_{2y})$ follows analogously.

For the last step (3y to 12y IRS), the new pseudo-discount pillars are, for each annual maturity t_i with $i = 3, \dots, 12$:

$$P(t_0, t_i) = \frac{P(t_0, t_{i-1})}{(1 + \tilde{\delta}_{2i-1} F_{2i-1})(1 + \tilde{\delta}_{2i} F_{2i})} \quad (8)$$

where the two factors in the denominator correspond to the two semi-annual floating periods of year i . Details are available in Baviera and Cassaro [2] to see how F_{2i} are computed.

2.d Multi-curve bootstrap results

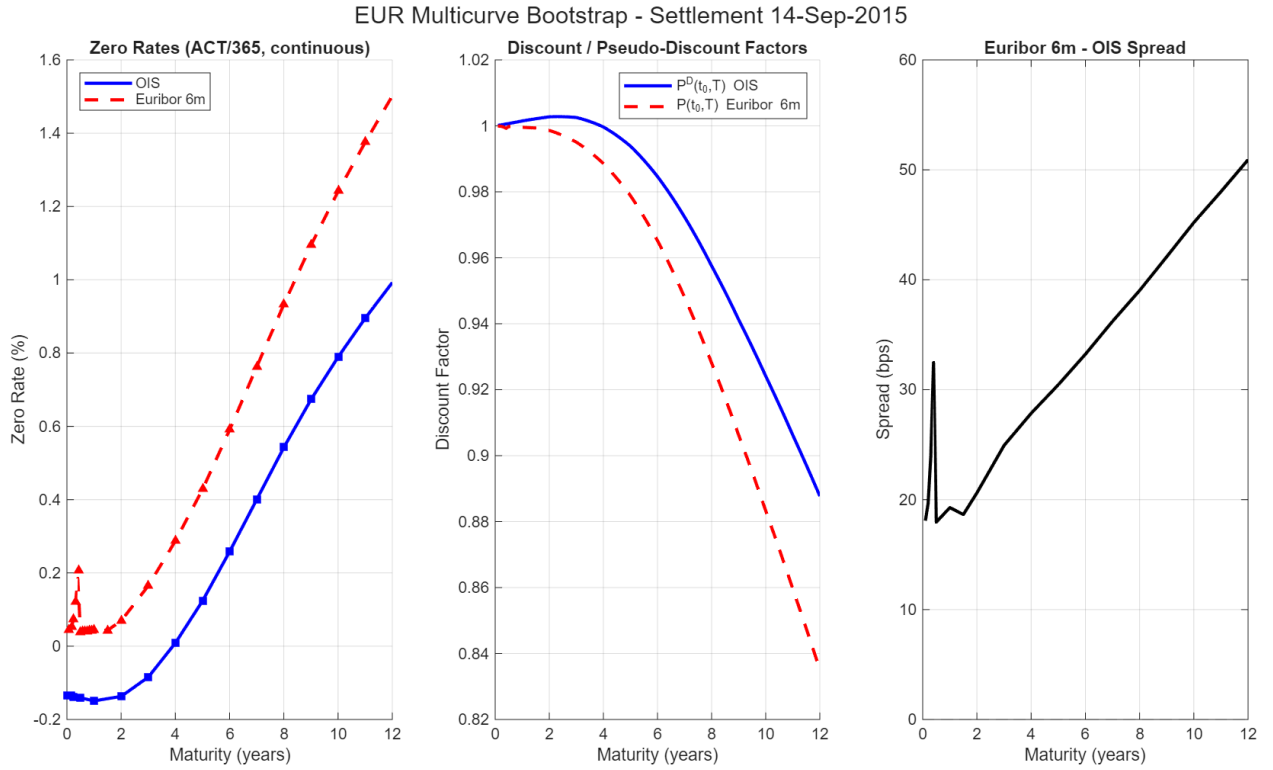


Figure 1: EUR multi-curve bootstrap at settlement date 14-Sep-2015.

Left: continuously-compounded zero rates (ACT/365) for the OIS and Euribor 6m curves, with the calibration pillars highlighted. **Centre:** OIS discount factors $P^D(t_0, T)$ and Euribor 6m pseudo-discount factors $P(t_0, T)$. **Right:** Euribor 6m–OIS spread in basis points.

Figure 1 summarises the output of the multi-curve bootstrap at the settlement date (14-Sep-2015), obtained by jointly calibrating the OIS discount curve and the Euribor 6m forward curve. The three panels show, respectively: the continuously-compounded zero rates (ACT/365 convention), the discount factors (and pseudo-discount factors for the Euribor 6m curve), and the Euribor 6m–OIS spread.

The OIS curve exhibits negative rates at the short end, while the Euribor 6m curve lies systematically above the OIS curve. As expected in a negative-rate area, discount factors are slightly above 1: for example, $P^D(t_0, T_{1w}) = 1.00002567$, corresponding to a zero rate of -0.1338% .

The OIS curve crosses below 1 around the 4 year mark, where rates turn positive. The difference between the two curves — the Euribor 6m–OIS basis, reported in the third panel — is positive and increasing with maturity, ranging from about 18 bps at the short end up to roughly 51 bps at 12 years. This spread reflects the credit and liquidity risk premium embedded in the Euribor rate relative to the (nearly) risk-free OIS rate, and provides the quantitative basis for the liquidity-premium analysis carried out in the following sections.

The Euribor 6m pseudo-discount falls to $P(t_0, t_{12y}) = 0.8349$ at 12 years, compared to 0.8876 for the OIS curve – a difference of more than 5 price points on a zero-coupon basis.

3 Volatility Calibration

We price the liquidity premium with a single, parsimonious dynamic model for the joint evolution of the discount and the spread curves: the three-parameter multi-curve Hull–White (MHW) model of Baviera [1], the minimal non-trivial extension of the classical two-parameter Hull–White to a multi-curve setting. Calibration follows the standard market cascade: the OIS discount curve $B(t_0, \cdot) \equiv P^D(t_0, \cdot)$ and the Euribor-6m pseudo-discount curve $\tilde{B}(t_0, \cdot) \equiv P(t_0, \cdot)$ are bootstrapped first from the liquid linear instruments (Section 2), and the three volatility parameters $p := (a, \sigma, \gamma)$ are then calibrated on a strip of swaptions. Fixing the curves first reflects the liquidity hierarchy of the instruments and is also numerically stable.

In the Gaussian multi-curve HJM framework the discount and the spread volatilities are

$$\sigma(t, T) = (1 - \gamma) \tilde{\sigma}(t, T), \quad \eta(t, T) = \gamma \tilde{\sigma}(t, T) \quad \tilde{\sigma}(t, T) = \begin{cases} \sigma \frac{1 - e^{-a(T-t)}}{a} & \text{if } a > 0 \\ \sigma(T - t) & \text{if } a = 0 \end{cases}. \quad (9)$$

The parameters have a clean reading: a is the mean-reversion speed (it sets how fast the volatility term structure decays with maturity), σ is the overall volatility level, and $\gamma \in [0, 1]$ is the fraction of the total volatility assigned to the spread ($\gamma = 0$ is a constant-spread model, $\gamma = 1$ the model of Baviera and Cassaro [2]). Crucially, the pseudo-discount volatility $\tilde{\sigma} = \sigma(t, T) + \eta(t, T)$ does not depend on γ , since $(1 - \gamma) + \gamma = 1$: γ only splits the total volatility between the discount and the spread legs, it does not change its magnitude.

We calibrate on the $M = 9$ co-terminal ATM cash-settled (CS) receiver swaptions of the 10y diagonal ($1y \times 9y, \dots, 9y \times 1y$), all versus Euribor 6m, with annual fixed leg (30/360) and semi-annual floating leg (Act/360). ATM options are the most liquid and the least affected by the smile, which a one-factor model does not aim to reproduce; CS swaptions are the EUR quoting standard, and their payoff uses the cash annuity (12) rather than the model BPV, which is why a dedicated pricing formula is required.

3.a Market prices: the Bachelier (Normal-Black) formula

The market quotes are turned into prices via the Normal-Black (Bachelier) formula for a CS receiver swaption,

$$R^{C, \text{mkt}} = B(t_0, t_\alpha) C(S_0) \left[(K - S_0) N(-d) + \sigma_N \sqrt{t_\alpha - t_0} \varphi(d) \right], \quad d = \frac{S_0 - K}{\sigma_N \sqrt{t_\alpha - t_0}}. \quad (10)$$

We use the normal rather than the lognormal (Black) convention because in September 2015 EUR rates were very low and negative at the short end. Black is ill-defined for negative or near-zero rates, the normal model handles them naturally, and the market itself quotes these swaptions in normal (basis-point) volatilities. The two multi-curve inputs are the ATM forward swap rate and the basis-point value,

$$S_0 = \frac{N_{\text{float}}}{\text{BPV}}, \quad \text{BPV} = \sum_j \delta_{j-1} B(t_0, t_j), \quad N_{\text{float}} = \sum_{\iota} \left(\frac{\tilde{B}(t_0, t_{\iota-1})}{\tilde{B}(t_0, t_{\iota})} - 1 \right) B(t_0, t_{\iota}), \quad (11)$$

where the forward pseudo-discount ratios in N_{float} carry the multi-curve (spread) correction. The par-yield cash annuity is

$$C(S) = \sum_{i=1}^{\omega-\alpha} \frac{1/m}{(1+S/m)^i} = \frac{1}{S} \left[1 - (1+S/m)^{-(\omega-\alpha)} \right], \quad S > -m, \quad (12)$$

with $m = 1$ payment per year in the EUR market, so $C(0) = \omega - \alpha$.

3.b Model prices: the closed-form CS swaption

The MHW model admits a semi-closed swaption formula extending the Jamshidian decomposition to the multi-curve case. The forward swap rate at expiry is written as a deterministic function $S(x)$ of a single standard Gaussian factor $x \sim \mathcal{N}(0, 1)$. With the accumulated standard deviation and the volatility coefficients

$$\zeta_{\alpha}^2 = \begin{cases} \sigma^2 \frac{1 - e^{-2a(t_{\alpha}-t_0)}}{2a} & \text{se } a > 0 \\ \sigma^2 (t_{\alpha} - t_0) & \text{se } a = 0 \end{cases}, \quad v_{\alpha,i} = \zeta_{\alpha} \frac{1 - e^{-a\tau_i}}{a}, \quad \tau_i = t_i - t_{\alpha}, \quad (13)$$

the discount- and spread-leg coefficients are

$$\xi_i = (1 - \gamma) v_{\alpha,i}, \quad \nu_i = v_{\alpha,i} - \gamma v_{\alpha,i+1}, \quad (14)$$

which is exactly where γ enters the pricing (ξ is the ς of the paper; the ν_i may change sign and are “extended” volatilities). The swap rate function reads

$$S(x) = \frac{N(x)}{\text{BPV}(x)}, \quad \text{BPV}(x) = \sum_j \delta_{j-1} B_j^{\alpha} e^{-\xi_j x - \frac{1}{2} \xi_j^2}, \quad (15)$$

$$N(x) = \sum_k \beta_{B,k} e^{-\nu_k x - \frac{1}{2} \nu_k^2} - \sum_k B_{k+1}^{\alpha'} e^{-\xi_{k+1} x - \frac{1}{2} \xi_{k+1}^2}, \quad (16)$$

where $B_j^{\alpha}, B_k^{\alpha'}$ are forward OIS discount factors and $\beta_{B,k} = \beta_{\iota} B_{\iota}^{\alpha'}$ collects the forward spread times the floating-leg forward discount. Since $S(\cdot)$ crosses the strike only once there is a unique critical value x^* with $S(x^*) = K = S_0$, and the receiver is in the money for $x < x^*$. The CS price is the one-dimensional Gaussian integral

$$R^{C,\text{MHW}} = B(t_0, t_{\alpha}) \int_{-\infty}^{x^*} \varphi(x) C(S(x)) [K - S(x)]^+ dx. \quad (17)$$

In the implementation x^* is bracketed on a coarse grid and refined with **fzero**, and the integral is evaluated by composite Simpson’s rule on 400 sub-intervals over $[-8, x^*]$.

The parameters are estimated by non-linear least squares on prices,

$$\hat{p} = \arg \min_p \text{Err}^2(p), \quad \text{Err}^2(p) = \sum_{i=1}^9 [R_i^{C,\text{MHW}}(p) - R_i^{C,\text{mkt}}]^2. \quad (18)$$

Matching prices rather than vols avoids an inner implied-vol inversion at every step and keeps the objective smooth. We minimise with `fmincon` (interior-point) under the box constraints $a \in [10^{-4}, 1]$, $\sigma \in [10^{-4}, 0.10]$, $\gamma \in [0, 1]$, using a four-point multistart to guard against local minima; all starts converge to the same optimum ($\text{Err} \approx 2.2 \times 10^{-3}$, exit flag 2).

3.c Results and interpretation

The optimiser converges, stably across all starting points, to

$$\hat{a} = 12.16\%, \quad \hat{\sigma} = 1.22\%, \quad \hat{\gamma} = 0.037\%, \quad (19)$$

with an aggregate price RMSE of 7.30×10^{-4} and a volatility RMSE of 2.03 bps.

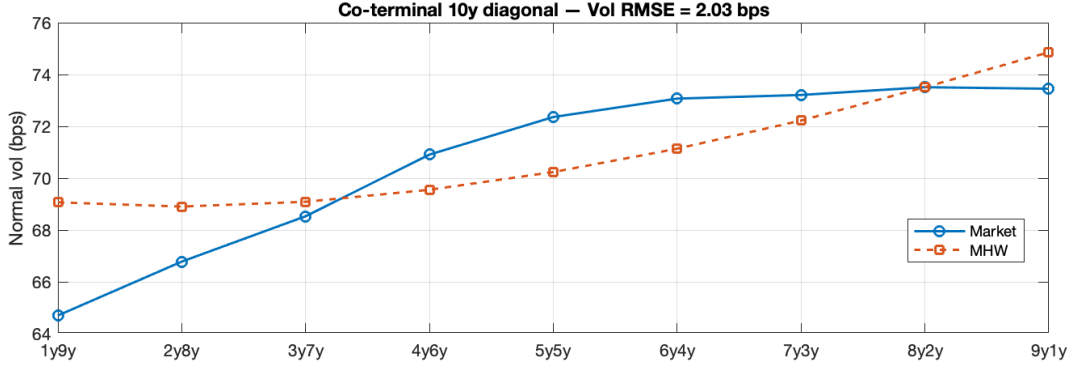


Figure 2: Market vs. MHW model ATM normal volatilities along the co-terminal 10y diagonal. The model captures the average level and the broad upward slope, but its profile is flatter than the market’s humped shape.

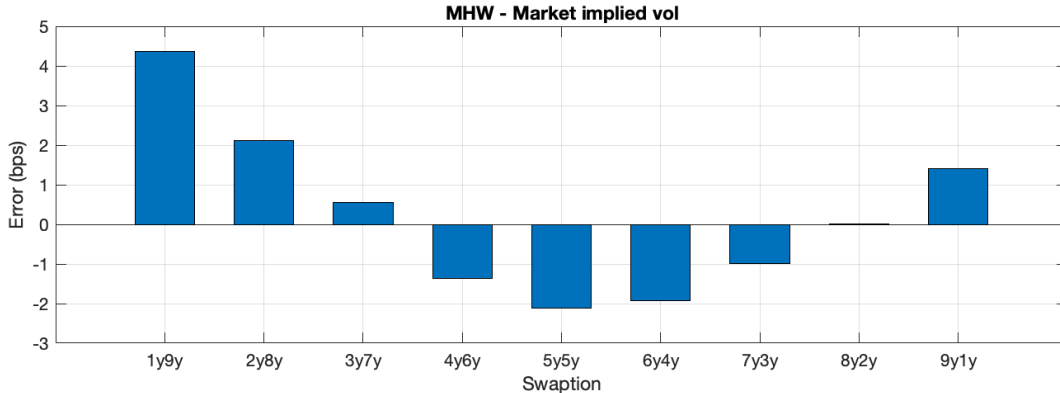


Figure 3: Per-swaption implied-volatility error (model – market) in basis points. The residuals oscillate around zero within a ± 4.4 bps band.

Figure 2 compares the model and market ATM normal volatilities along the diagonal, and Figure 3 reports the per-swaption errors. These values are close to, but not identical to, the references in

Baviera [1] ($a = 12.94\%$, $\sigma = 1.26\%$, $\gamma = 0.07\%$), and the small gap is expected, since the published calibration may use a different optimiser, it relies on three FRAs only, whereas our Euribor 6m curve also uses an 18-month swap rate that is not directly quoted but obtained by interpolation.

A volatility RMSE of about 2 bps against market vols of 65–75 bps is a relative error of roughly 3% – a very good result for a three-parameter model fitted to nine instruments. The errors (Figure 3) are positive at the two ends of the diagonal and negative in the belly: this is the typical signature of a one-factor model, whose volatility term structure is necessarily smoother than the market’s humped profile (Figure 2), so the least-squares fit trades a slight over-estimation at the wings against a slight under-estimation in the middle. As for the parameters, a and σ are well determined – together they fix the pseudo-discount volatility $\tilde{\sigma}$ – while γ , the fraction of $\tilde{\sigma}$ carried by the spread ($\eta = \gamma\tilde{\sigma}$), is only weakly identified: the near-zero estimate ($\hat{\gamma} \approx 0.04\%$) reflects the fact that diagonal ATM swaptions, and the absence of liquid OIS options, cannot resolve the discount/spread split. This is benign downstream: the Section 4 bounds depend only on $\hat{a}$ and $\hat{\sigma}$ through the cumulated volatility (24), so the computed premium is independent of how the volatility is split.

4 Bounds of the liquidity premium

4.a Methodology

The objective of this section is to quantify the *sheer liquidity premium* Δ_τ , defined as the price difference between an otherwise identical liquid and illiquid defaultable coupon bond:

$$\Delta_\tau = \bar{P}(t_0, c, \mathbf{t}) - \bar{\bar{P}}_\tau(t_0, c, \mathbf{t}) \quad (20)$$

where $\bar{P}$ denotes the liquid bond price and $\bar{\bar{P}}_\tau$ the illiquid bond price with time-to-liquidate τ . We consider two values of τ : two weeks and two months from the value date $t_0 = 10$ September 2015. For this part, we will need three main ingredients. First the bootstrapped OIS discount curve $P^D(t_0, T)$, which provides risk-free discount factors $B(t_0, t_i)$ at each bond payment date. Second, the bootstrapped Zeta-spread curve $Z(T)$: combined with the OIS curve, it gives the defaultable zero-coupon bond price $\bar{B}(t_0, T) = B(t_0, T) e^{-Z(T)(T-t_0)}$. And finally, the volatility parameters $\hat{a}$ and $\hat{\sigma}$.

We consider zero recovery throughout. The computation is performed bond by bond for all liquid benchmark issues of BNP Paribas (8 bonds) and Santander (9 bonds), for each of the two TTL values.

4.b Zeta Spread Bootstrap

Before applying the bounds we summarise each issuer’s credit information through the Zeta-spread curve $Z(T)$. For every issuer we bootstrap Z from its liquid benchmark bonds, sorted by increasing maturity, requiring that the model dirty price reproduce the market invoice (dirty) price—clean price plus accrued interest, the latter computed Act/Act ICMA—of each bond.

The defaultable discount factor at a payment date t is $\bar{B}(t_0, t) = P^D(t_0, t) e^{-Z(t)(t-t_0)}$, with P^D the OIS discount factor and $Z(t)$ the continuously-compounded Z-spread (Act/365). For a bond with cash flows c_i at dates t_i , the model dirty price reads

$$\bar{P}(t_0) = \sum_i c_i P^D(t_0, t_i) e^{-Z(t_i)(t_i-t_0)}. \quad (21)$$

The bootstrap is sequential, with curve pillars equal to the bond maturities $T_1 < \dots < T_N$. At step k the spreads at the previous pillars $T_1, \dots, T_{k-1}$ are already known: the cash flows of bond k falling on or before T_{k-1} are discounted with the Z-spread linearly interpolated on those pillars (known present value). The remaining cash flows, between T_{k-1} and T_k , use a spread that varies linearly in time from $Z(T_{k-1})$ to the unknown $z = Z(T_k)$:

$$Z(t_i) = Z(T_{k-1}) + (z - Z(T_{k-1})) w_i, \quad w_i = \frac{t_i - T_{k-1}}{T_k - T_{k-1}}. \quad (22)$$

The pillar value $z = Z(T_k)$ solves

$$\underbrace{\sum_{t_i \leq T_{k-1}} c_i \bar{B}(t_0, t_i)}_{\text{known PV}} + \sum_{t_i > T_{k-1}} c_i \bar{B}(t_0, t_i; z) = \bar{P}_k^{\text{mkt}},$$

found with a bracketed root search on $[-0.01, 0.50]$.

Table 1 reports the bootstrapped Z-spreads for the two issuers, together with the time to maturity τ of each liquid benchmark bond.

Table 1: Bootstrapped Z-spreads $Z(T_k)$ (continuously compounded, Act/365) for the BNP Paribas and Santander benchmark bonds.

BNP Paribas			Santander		
Maturity	τ (y)	Z (bp)	Maturity	τ (y)	Z (bp)
27-Nov-2017	2.2055	45.60	27-Mar-2017	1.5342	61.70
12-Mar-2018	2.4932	49.41	04-Oct-2017	2.0575	65.42
21-Nov-2018	3.1890	63.38	15-Jan-2018	2.3397	67.30
28-Jan-2019	3.3753	68.99	20-Apr-2018	2.6000	77.34
23-Aug-2019	3.9425	71.70	14-Jan-2019	3.3370	83.98
13-Jan-2021	5.3370	91.72	13-Jan-2020	4.3342	94.59
24-Oct-2022	7.1151	96.81	24-Jan-2020	4.3644	94.80
20-May-2024	8.6877	103.44	14-Jan-2022	6.3397	112.79
			10-Mar-2025	9.4932	118.85

The Zeta-spread is the issuer's credit component of the bond spread: under the zero-recovery assumption it encodes the whole default-risk information, and each curve is taken constant up to the shortest maturity and linearly interpolated afterwards. For both issuers the bootstrapped spreads are positive and mildly increasing with maturity, of the order of a few tens of basis points, and define the defaultable discount factors $\bar{B}(t_0, T_k) = P^D(t_0, T_k) e^{-Z(T_k)(T_k - t_0)}$ used in the liquidity-premium bounds of next section.

4.c Lower and upper bounds

The main result of paper [3] (equation (13)) states, using a lower and upper bound :

$$\sum_{\substack{i=1 \\ t_i > \tau}}^N c_i \bar{B}(t_0, t_i) \left(\pi_i^L(\tau) - P(t_0, \tau) \right) \leq \Delta_\tau \leq \sum_{\substack{i=1 \\ t_i > \tau}}^N c_i \bar{B}(t_0, t_i) \left(\pi_i^U(\tau) - P(t_0, \tau) \right) \quad (23)$$

where the sum runs over all payment dates $t_i > \tau$, c_i are the bond cash flows (coupons plus face value at maturity), $\bar{B}(t_0, t_i)$ are the defaultable (risky) discount factors, and $P(t_0, \tau)$ is the issuer survival probability up to τ .

The quantities $\pi_i^L(\tau)$ and $\pi_i^U(\tau)$ depend on the *cumulated volatility* $\Sigma_i(\tau)$, which measures the integrated variance of the forward defaultable zero-coupon bond over the liquidation horizon:

$$\Sigma_i(\tau)^2 = \xi_i^2 \cdot \frac{1 - e^{-2\hat{a}\tau}}{2\hat{a}}, \quad \xi_i = \frac{\hat{\sigma}}{\hat{a}} \left(1 - e^{-\hat{a}(t_i - \tau)}\right) \quad (24)$$

with τ and $t_i - \tau$ both expressed in Act/365 year fractions.

The upper bound uses the closed-form expression for the expected maximum of a driftless geometric Brownian motion:

$$\pi_i^U(\tau) = \frac{4 + \Sigma_i^2}{2} \mathcal{N}\left(\frac{\Sigma_i}{2}\right) + \frac{\Sigma_i}{\sqrt{2\pi}} e^{-\Sigma_i^2/8} \quad (25)$$

The lower bound involves a one-dimensional integral over $\eta \in (0, 1)$:

$$\pi_i^L(\tau) = \frac{e^{-\Sigma_N^2/8}}{\pi} \int_0^1 \frac{A(\eta) B(\eta) C(\eta)}{\sqrt{\eta(1-\eta)}} d\eta \quad (26)$$

where $\Sigma_N = \Sigma_N(\tau)$ is the cumulated volatility computed at the bond maturity t_N , and the functions A , B , C are:

$$\begin{aligned} A(\eta) &= \exp\left(-\frac{\eta}{2} \Sigma_i(\Sigma_i - \Sigma_N)\right) \\ B(\eta) &= 1 + \sqrt{\frac{\pi(1-\eta)}{2}} \Sigma_N e^{(1-\eta)\Sigma_N^2/8} \mathcal{N}\left(\frac{\sqrt{1-\eta}}{2} \Sigma_N\right) \\ C(\eta) &= 1 + \sqrt{\frac{\pi\eta}{2}} (2\Sigma_i - \Sigma_N) e^{\eta(2\Sigma_i - \Sigma_N)^2/8} \mathcal{N}\left(\frac{\sqrt{\eta}}{2} (2\Sigma_i - \Sigma_N)\right) \end{aligned}$$

The integrand has integrable singularities at $\eta = 0$ and $\eta = 1$; numerical integration is performed with tight tolerances and intermediate waypoints to ensure accuracy.

The difference $\Delta_\tau^U - \Delta_\tau^L$ is the central output of this section. From (23), the survival probability $P(t_0, \tau)$ cancels exactly in the difference:

$$\Delta_\tau^U - \Delta_\tau^L = \sum_{\substack{i=1 \\ t_i > \tau}}^N c_i \bar{B}(t_0, t_i) (\pi_i^U(\tau) - \pi_i^L(\tau)) \quad (27)$$

This quantity measures the precision of the approximation: the paper shows that it is negligible for all practical purposes, confirming that either bound provides a reliable closed-form estimate of the true liquidity premium Δ_τ .

Figure 4 shows the difference $\Delta_\tau^U - \Delta_\tau^L$ as a function of bond maturity, for the two issuers and the two times-to-liquidate.

For every bond the difference is positive and extremely small—at most of order 10^{-8} per unit face value—and grows both with maturity and with the time-to-liquidate. Its negligible size confirms that the two bounds effectively coincide, so either one can be taken as the value of the sheer liquidity premium.

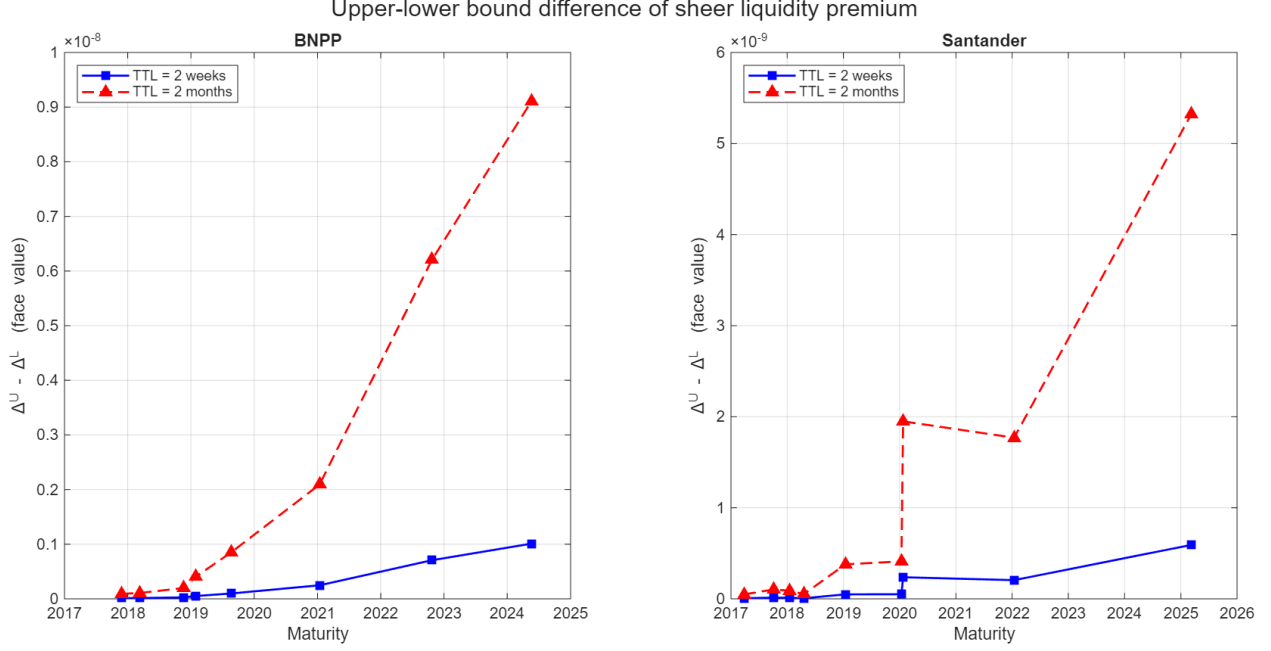


Figure 4: Difference between the upper and lower bound of the sheer liquidity premium, $\Delta^U_\tau - \Delta^L_\tau$ (per unit face value), versus maturity, for BNP Paribas (left) and Santander (right), with τ equal to two weeks and two months. Note the different vertical scales (10^{-8} and 10^{-9}).

5 Bond yields

The sheer liquidity premium of Section 4.c is expressed in price units. To report it in the metric used by practitioners, and to make it comparable across bonds of different coupon and maturity, we translate it into a yield spread. Throughout we use continuous compounding and Act/365 year fractions, consistent with the zero-rate convention adopted for the curves in Section 2.

Liquid yield. The liquid (dirty) price $\bar{P}$ of a bond is its model price of Section 4.b, equal by construction to the market invoice price,

$$\bar{P} = \sum_i c_i \bar{B}(t_0, t_i), \quad (28)$$

where $\bar{B}(t_0, t_i) = P^D(t_0, t_i) e^{-Z(t_i)(t_i - t_0)}$ is the defaultable discount factor introduced in Section 2. The liquid yield Y is the internal rate of return that reproduces $\bar{P}$ from the same cash flows under a single, flat, continuously-compounded rate,

$$\bar{P} = \sum_i c_i e^{-Y(t_i - t_0)}. \quad (29)$$

Unlike the term structure $\{\bar{B}(t_0, t_i)\}$, the yield Y is a scalar summary of the whole cash-flow stream; (29) is solved bond by bond with a bracketed root search on $[-0.05, 0.30]$, an interval wide enough to contain the (possibly negative) EUR yields of September 2015.

Illiquid price. The illiquid price at time-to-liquidate τ subtracts the premium from the liquid price,

$$\bar{\bar{P}}_\tau = \bar{P} - \Delta^U_\tau. \quad (30)$$

We use the upper bound Δ_τ^U for two reasons: the two bounds coincide to order 10^{-8} , so the choice is immaterial in value; and Δ_τ^U is available in closed form, avoiding the one-dimensional integral required by the lower bound. No cash flow is dropped at this stage: the truncation at the liquidation date is already embedded in Δ_τ^U , so $\bar{P}$ and $\bar{\bar{P}}_\tau$ are discounted over the identical cash-flow set.

Illiquid yield and liquidity spread. The illiquid yield y_τ solves the same internal-rate-of-return equation with the illiquid price,

$$\bar{\bar{P}}_\tau = \sum_i c_i e^{-y_\tau (t_i - t_0)}. \quad (31)$$

Writing $\ell_\tau := y_\tau - Y$ for the continuously-compounded liquidity component, the illiquid price decomposes as

$$\bar{\bar{P}}_\tau = \sum_i c_i e^{-(Y + \ell_\tau)(t_i - t_0)}, \quad (32)$$

so the bond yield splits into a default part—carried by Y through the Zeta-spread—and a liquidity part. We report the latter in basis points,

$$\tilde{L}_\tau = 10^4 \ell_\tau = 10^4 (y_\tau - Y). \quad (33)$$

Since $\bar{\bar{P}}_\tau < \bar{P}$ we have $y_\tau > Y$ and hence $\tilde{L}_\tau > 0$; the spread also widens with τ , the longer liquidation horizon exposing the holder to default risk over a longer window. The procedure is applied to both issuers, all of their benchmark bonds, and the two values of τ (two weeks and two months).

It is worth stressing that we deliberately measure liquidity at coupon-bond level rather than through the per-zero-coupon spread of [3, Eq. (17)]. That formula assigns a separate liquidity spread to each cash-flow date t_i , i.e. a whole term structure $\{L_\tau(t_i)\}_i$; a coupon bond, however, is quoted on a single yield, its internal rate of return over all cash flows, so its liquidity component is necessarily one scalar per bond. Equation (33) delivers that scalar directly—by solving the same IRR on the illiquid price $\bar{\bar{P}}_\tau$ —which is precisely the quantity plotted in Figures 4–5 and comparable across bonds of different coupon and maturity. Using (33) therefore avoids computing a per-maturity spread that would have to be re-aggregated into a single bond yield anyway.

Figure 5 shows, for each issuer, the liquid yield together with the two illiquid yields (time-to-liquidate of two weeks and two months) against bond maturity.

For both issuers the liquid yield curve is upward sloping in maturity (from about 0.33% to 1.61% for BNP Paribas and from 0.47% to 1.89% for Santander). The two illiquid curves lie above the liquid one and are ordered by horizon—the two-month curve above the two-week one—reflecting the strictly positive liquidity premium; the gap, however, is small relative to the yield level, of the order of a few tens of basis points.

Table 2 reports the liquidity yield spread $\tilde{L}_\tau$ in basis points for the two horizons.

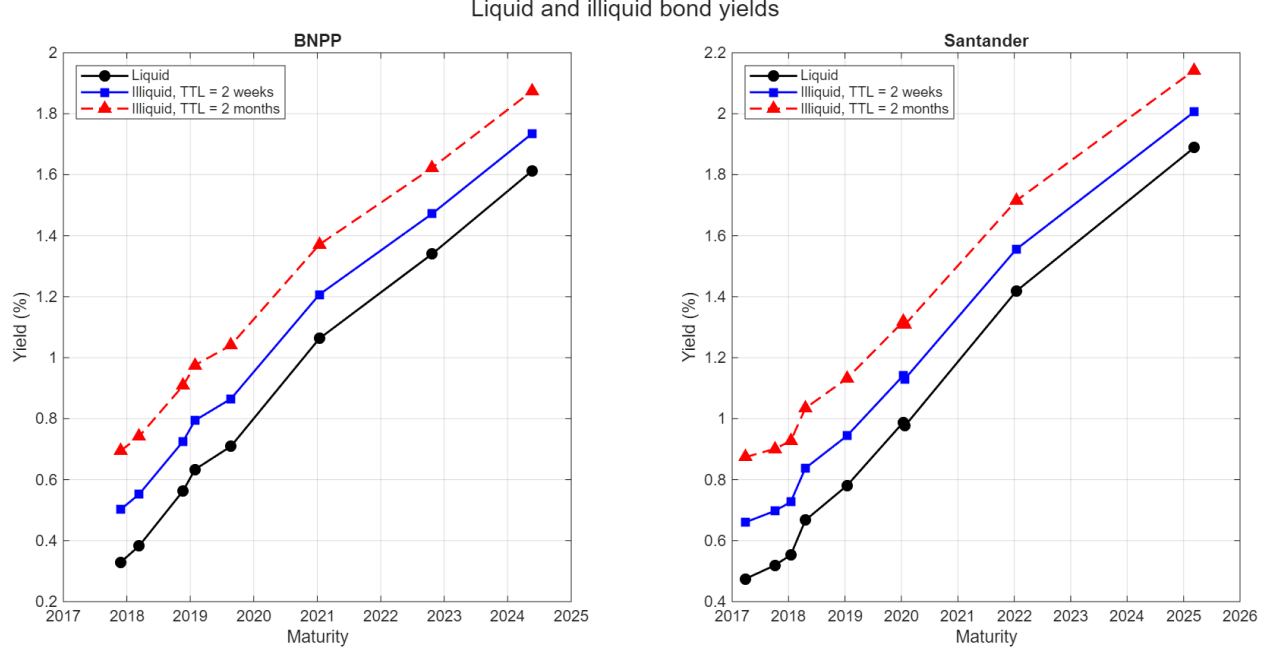


Figure 5: Liquid and illiquid bond yields versus maturity, for BNP Paribas (left) and Santander (right), with τ equal to two weeks and two months.

Table 2: Liquidity yield spread $\tilde{L}_\tau$ (basis points) for the BNP Paribas and Santander benchmark bonds, at τ equal to two weeks and two months.

BNP Paribas			Santander		
Maturity	$\tilde{L}_{2w}$ (bp)	$\tilde{L}_{2m}$ (bp)	Maturity	$\tilde{L}_{2w}$ (bp)	$\tilde{L}_{2m}$ (bp)
27-Nov-2017	17.3122	36.5384	27-Mar-2017	18.5779	39.9380
12-Mar-2018	16.9713	35.9028	04-Oct-2017	17.8188	38.2000
21-Nov-2018	16.2553	34.5110	15-Jan-2018	17.4156	37.4534
28-Jan-2019	16.0882	34.1791	20-Apr-2018	17.0830	36.7455
23-Aug-2019	15.5767	33.1589	14-Jan-2019	16.3176	35.1229
13-Jan-2021	14.4297	30.8222	13-Jan-2020	15.3437	33.0583
24-Oct-2022	13.2038	28.2588	24-Jan-2020	15.4148	33.2109
20-May-2024	12.2124	26.2118	14-Jan-2022	13.7420	29.6596
			10-Mar-2025	11.7292	25.3700

The liquidity yield spread is positive for every bond and at every horizon, ranging from about 12 to 19 bps at two weeks and from 25 to 40 bps at two months. Two regularities stand out. First, the spread grows sub-linearly with the time-to-liquidate: two months is roughly 4.3 times two weeks in calendar time, yet the spread only about doubles ($\approx 2.1\times$), consistent with the $\sqrt{\tau}$ growth of the cumulated volatility that drives the premium. Second, at a given horizon the spread declines with maturity, as the same price premium translates into a smaller yield adjustment when spread over the longer duration of a longer-dated bond. The magnitudes are broadly similar across the two issuers.

6 Conclusion

This study successfully quantified the corporate bond liquidity premium for BNP Paribas and Santander using the multi-curve framework as of September 2015. By bootstrapping the OIS and Euribor 6m curves, calibrating the multi-curve Hull–White model, and applying tight analytical bounds, we isolated the pure liquidity component from credit risk.

Our empirical results confirm two key market regularities. First, the liquidity yield spread grows sub-linearly with the time-to-liquidate ($\approx 2.1\times$ from two weeks to two months), consistent with the square-root-of-time growth of cumulated volatility. Second, for a given liquidation horizon, the spread decreases with bond maturity because the same price premium is amortized over a longer duration. Ultimately, the negligible difference of order 10^{-8} between the upper and lower bounds proves that the closed-form formulas provide a highly reliable and mathematically robust tool for practitioners to evaluate liquidity risk in fixed-income markets.

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